

Automating the Creation and Placement of Objects in a Scene using Python Scripting in Autodesk Maya

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ABSTRACT

This hands on lab will use Python scripting in Maya to automate the creation and placement of objects in a scene.

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1 TECHNICAL DETAILS

We will complete necessary steps to set up communication between Python and Autodesk Maya.

1.1 Creating Objects

We will show attendees the code to create primitives. Listing 1 illustrates the use of a loop to draw five spheres. We would use nested loops to show grids of objects. Attendee would be guided to create grids of different dimensions using different primitive objects.

```
1 # for loops in maya
2 # this program works the same as listing 6.8
3 import maya.cmds as cmds
4
5 cmds.file( force = True, new = True) # create a new
  file
6
7 sphereCount = 10 # this can be changed easily
8 xTranslate = 2 # as can this
9
10 for eachSphere in range(sphereCount): # runs from 0 to 9
    so 10 spheres
11     sphere = cmds.polySphere()
12     cmds.move(xTranslate*eachSphere, 0, 0, sphere)
```

Listing 1: Using a loop create and place spheres

1.2 The Random Package

In Listing 2 we can place the cubes on a grid and then scale them a random amount in y using the random package. We would use random to change other variables also including size, position, orientation, and color (as shown in Figure 1).

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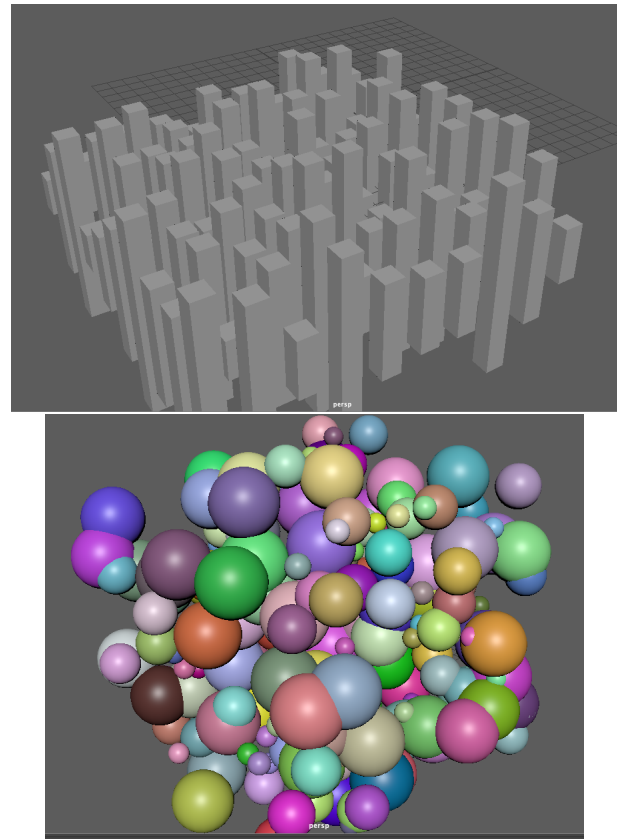


Figure 1: Using the random package to generate objects of different scale, and different colors

```
1 import maya.cmds as cmds
2 import random
3 cmds.file( force = True, new = True) # create a new
  file
4
5 # using nested loops to make a grid of objects
6 gridX = 10
7 xTranslate = 2
8
9 gridY = 10
10 yTranslate = 2
11
12 for xElem in range(gridX): # runs from 0 to 9 so 10
    spheres
13     x = xTranslate * xElem
14     for yElem in range(gridY): # runs from 0 to 9 so 10
        spheres
```

```
15         z = yTranslate * yElem
16         cube = cmds.polyCube()
17         cmds.move(x, 0, z, cube)
18
19 geoList = cmds.ls( geometry=True )
20 print geoList
21
22 # this will scale everything by a random amount in the y
   direction
23 # the for loop will iterate over each geo object
24 # and scale it
25 for geo in geoList:
26     cmds.scale(1, random.randint(2,10), 1, geo)
27
28 cmds.select(clear = True)
```

Listing 2: Using a for loop scale up all geometry in a scene

2 METHOD OF PRESENTATION

Traditional lab style with the content projected and the attendees following along on their own machines.

3 INTENDED AUDIENCE

Anyone who would like to automate repetitive tasks when working in Autodesk Maya, students, educators, hobbyists, and professionals.

4 EXPECTED TAKE AWAYS

By the end of the course, attendees should walk away the ability to write basic scripts for Maya. Hopefully, they will have the tools, confidence, and initiative to explore more advanced scripts independently.